

# Week 2 Discussion

**Covers:** the axioms and calculus of probabilities (Week 1, Lecture 1.2) and counting (Week 2, Lecture 2.1).

Work through each problem in steps. Hints are given; final answers are hidden behind “Show answer” so you can check your work.

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## Problem 1 (Working with the probability rules)

In a class, a student is chosen at random. Let  $R$  be the event that the student uses R and  $Y$  the event that the student uses Python, with

$$P(R) = 0.60, \quad P(Y) = 0.45, \quad P(R \cap Y) = 0.25.$$

For each answer below, name the rule you used (complement rule, Theorem 1.2.8; Theorem 1.2.9; Bonferroni’s inequality).

**(a)** Find the probability that the student uses

1. at least one of the two languages;
2. neither language;
3. R but not Python;
4. exactly one of the two languages.

*Hint:* write each event in set notation first, for example “R but not Python” is  $R \cap Y^c$ .

**(b)** Now suppose you are told only  $P(R) = 0.60$  and  $P(Y) = 0.45$ , but not  $P(R \cap Y)$ . What are the smallest and largest possible values of  $P(R \cap Y)$ ? Could  $R$  and  $Y$  be disjoint?

*Hint:* one bound comes from Bonferroni’s inequality, the other from monotonicity ( $R \cap Y \subset Y$ ).

**(c)** Show that both bounds in (b) can actually be attained. Take a class of 100 students, each equally likely to be chosen, and describe which students use each language.

► [Show answer](#)

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## Problem 2 (The matching problem)

Each of  $n$  letters is placed at random into one of  $n$  addressed envelopes, one letter per envelope, so all  $n!$  arrangements are equally likely. What is the probability that **at least one** letter goes into its correct envelope?

Let  $A_i$  be the event that letter  $i$  is in its correct envelope. We want  $P\left(\bigcup_{i=1}^n A_i\right)$ . The events overlap, so we use the general **inclusion–exclusion** formula, which extends the two- and three-set versions from

lecture:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

- (a)** Find  $P(A_i)$ . *Hint:* if letter  $i$  is fixed in its envelope, how many ways can the other  $n - 1$  letters be arranged?
- (b)** Find  $P(A_i \cap A_j)$  for  $i \neq j$ , and more generally  $P(A_{i_1} \cap \cdots \cap A_{i_k})$  for any  $k$  distinct letters.
- (c)** How many terms are in the  $k$ th sum  $\sum_{i_1 < \cdots < i_k}$ ? Show that the whole  $k$ th sum equals  $1/k!$ .
- (d)** Conclude that

$$P(\text{at least one match}) = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1} \frac{1}{n!}.$$

- (e)** Evaluate for  $n = 3$  and  $n = 4$ . Check  $n = 3$  by listing all  $3! = 6$  arrangements. What happens as  $n \rightarrow \infty$ ? *Hint:* recall  $e^x = \sum_{k=0}^{\infty} x^k/k!$ .

► *Show answer for (e)*

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### Problem 3 (Galileo's dice problem)

Around 1620, gamblers asked Galileo a puzzle: when three dice are rolled, a sum of **9** and a sum of **10** can each be made in six ways, yet experience showed that 10 comes up more often. Why?

- (a)** List the six unordered ways to write 9 as a sum of three dice values (for example  $\{1, 2, 6\}$ ), and do the same for 10.
- (b)** The gamblers' reasoning treats the unordered outcomes as equally likely. Which sample space is equally likely for three fair dice, and how many outcomes does it have?
- (c)** For each unordered outcome in (a), count the ordered outcomes that produce it. *Hint:* use the multinomial coefficient. For example,  $\{1, 4, 4\}$  corresponds to  $\frac{3!}{1!2!} = 3$  ordered rolls.
- (d)** Compute  $P(\text{sum} = 9)$  and  $P(\text{sum} = 10)$ .
- (e)** How many distinct *unordered* outcomes are there for three dice? Use the unordered-with-replacement formula from Table 1.2.1, and explain why this number cannot be used as  $N(S)$  in  $P(A) = N(A)/N(S)$ .

► *Show answer for (d) and (e)*